Machine Learning

Crash Course Notes

ML | DL | GenAI

8 Days | 42 Pages | April 2026

Innomatics Research Labs

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Day 1 13 April 2026 | Introduction to Data Analysis & Machine Learning

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1. Data Analysis (DA)

Data Analysis is the process of examining raw data to extract meaningful information.

- Raw Data → Unprocessed information (numbers, text, images)
- Examine raw data to get meaningful information
- DA tells us what happened; ML predicts what will happen

Example: A class with students A→80, B→90, C→85, D→95, E→82 → Summarize → Mean = 85 → Information

Data Analysis Workflow

- Collect data
- Organize → Clean data (Database Management)
- Analyze data (EDA → Univariate, Bivariate, Multivariate)
- Visualize (PowerBI, R, Python, MySQL)
- Get Information

EDA Tools: MySQL (Queries), Python, PowerBI | Focus: Salary, Profit data

Study Hours vs Score Example

Hidden pattern: Every 2 hrs of study → increase in score.

Study Hours (S-H)	Score
2	50
4	60
6	70
8	80
10	90

If a student studies 7 hours → predicted score ~65 (machine learns this pattern)

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2. Artificial Intelligence (AI)

AI = Ability of machines to learn, think, understand, and make decisions.

- Natural Intelligence (NI) → Produced by Brain (Humans & Animals)
- Brain → Intelligence → Action (think, learn, decide)
- AI → Given to Machine (via Data)
- Human intelligence: Learn → Understand → Think → Solve

How AI Works

- Human: touch fire → feels pain → avoid
- Machine: Data → AI System/App → New data → Proper Output
- Mobile → Face Unlock → Learn Features

3. Machine Learning (ML)

ML is a part of AI that allows machines to learn from data and make predictions/decisions without being explicitly programmed.

Key Concept

Machine Learning: Learns patterns from data → Uses those patterns → Makes future predictions

- DA → Tells us what happened
- ML → Predicts what will happen

How Does a Machine Learn?

- Give data to (Learn) → Ask Question → Output
- Machine learns hidden patterns and uses them to predict
- Give data → Algorithm → Output

AI vs ML Relationship

ML is a subset of AI. ML ⊂ AI

Example: AI → Chatbot → 'Hello' → 'Hello, how are you?' (trained on data + machine learning)

Types of Machine Learning

Type	Description	Labeled Data?
Supervised Learning	Model learns from labeled data	Yes
Unsupervised Learning	Model finds patterns without labels	No
Reinforcement Learning	Model learns by reward/punishment	No

How Does the Machine Learn from Data?

- Supervised Learning ✓
- Unsupervised Learning

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4. Supervised Learning

Model learns from labeled data (Input → Output pairs).

- Input: Independent variable (X)
- Output: Dependent variable (Y / Label)

Example: Student study hours (X) → Marks (Y) — labeled training data

Types of Supervised Learning

- Regression → Numerical (Continuous) data
- Classification → Categorical data

Data Type	Algorithm Type	Output Example
Numerical	Regression	Predict score: 75
Categorical	Classification	Predict: Pass/Fail, Yes/No

Regression

- Predict Numerical (Continuous) values
- Find relationship between variables
- Dependent variable (Target) → what we want to predict (Y)
- Independent variable (Predictor) → inputs used for prediction (X)

Example: S-H (X=Independent) | Score (Y=Dependent): 2→50, 4→60, 6→70, 8→80, 10→90 → Regression learns relationship between X and Y

Classification

- Predict Categories instead of numbers
- Output → Yes/No, T/F, Spam/Not Spam, Pass/Fail

Example: Marks→ Pass/Fail | CIBIL Score → Loan Approve: Yes/No (< 700 → No, ≥ 700 → Yes)

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5. Machine Learning Workflow

The end-to-end process to build and validate an ML model:

- 1. Collect Data
- 2. Data Cleaning & Preprocessing
- 3. Split Data → Train (80%) / Test (20%)
- 4. Train the Model (Regression / Classification Algorithm)
- 5. Test the model on test data
- 6. Output → Predicted value
- 7. Compare: Actual = Predicted → Machine learned successfully ✓
- Actual ≠ Predicted → Failed ✗

Training: Give labeled data to ML Algorithm → Learns patterns. Testing: New unseen data → Predict output → Compare with actual.

Day 2 14 April 2026 | Data Preprocessing

1. Data Preprocessing

Cleaning, organizing & transforming raw data into a format used in ML.

- Huge raw data → Machine → Wrong results if not preprocessed
- Proper format + encoding → Correct predictions

If Actual=100 and Predicted=75 → Error detected → Fix via preprocessing (nearest to actual value)

Steps of Data Preprocessing

1) Data Cleaning

- Handle missing values → Replace with Mean/Median/Mode, Remove rows, Fill manually
- Remove duplicates
- Fix errors

Example: Study hours 2,3,4 but Score has Null,70,80,90 → Replace Null with average

2) Data Transformation

- Convert data into usable format
- Text → Numbers (via Encoding)
- Gender: Male/Female → Machine needs numbers → Encoding

3) Data Scaling (Normalization & Standardization)

Different columns have different ranges → Need to bring to same range → Avoids bias.

Example: Age (12–16) vs Salary (10k–50k) → Biased toward Salary → Apply Scaling → Same range

4) Encoding

Machines understand only numbers. Categorical text must be converted.

- Why? → All ML algorithms work only with numeric data

Types of Encoding

a) Label Encoding

- Assign unique number to each category
- No duplicate numbers

Color: Red→0, Green→1, Blue→2, White→3 (red repeats → gets same 0)

b) One Hot Encoding (OHE)

- Creates separate columns for each category
- Binary values (0/1) for each column

Color	Red	Blue
Red	1	0
Blue	0	1

c) Ordinal Encoding (OE)

- Assign numbers based on order/ranking

Size: Small→1, Medium→2, Large→3 (ordered by size)

5) Scaling

Process of bringing all numerical data into a similar range.

i) Normalization (Min-Max Scaling)

Scale data into fixed range (0 to 1).

```
X_norm = (X - X_min) / (X_max - X_min)
```

Example: Marks 20,40,60,80,100 → Min=20, Max=100 → A=(20-20)/80=0, B=0.25, C=0.5, D=0.75, E=1.0

ii) Standardization (Z-Score Normalization)

Centers data around mean with standard deviation.

```
Z = (X - μ) / σ
```

where: μ = mean, σ = standard deviation

Example: Marks 10,20,30,40,50 → Mean=30, σ=14.14 → A=(10-30)/14.14=-1.41 → values centered around 0 (-ve and +ve)

6) Splitting Data

- Train data → Machine learns patterns
- Test data → Evaluate the model

Typical split: 80% Train, 20% Test

5) Outliers

- Extremely high or low values that are inconsistent
- Example: Salaries 10k, 12k, 14k, 16k, 2 lakhs → 2 lakhs is outlier
- Options: Remove or Keep (depending on context)

Data Preprocessing Summary

Step	Technique	Purpose
Data Cleaning	Handle nulls, remove duplicates, fix errors	Remove noise
Data Transformation	Encoding (Label, OHE, Ordinal)	Convert text to numbers
Data Scaling	Normalization, Standardization	Bring to same range
Outlier Handling	Remove or keep outliers	Improve consistency
Splitting	Train / Test split	Model training & evaluation

Day 3 16 April 2026 | Regression Algorithms

ML Pipeline Recap

Collect Data → Data Preprocessing (DP) → Algorithm (Training) → New Data → Output

- Supervised: Regression ✓ / Classification ✓
- Unsupervised
- Reinforcement

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1. Regression Algorithm

Used to predict numerical/continuous values.

- Find relationship between input (X) and output (Y)
- Use Regression in ML → Train machine using Regression

Types of Regression

- Linear Regression ← very important
- Multiple Linear Regression
- Polynomial Linear Regression
- Ridge Regression
- Lasso Regression

Linear Regression (LR)

When the relationship between input and output follows a straight line pattern.

- X → Independent variable
- Y → Dependent variable
- LR draws a straight line (best fit line) that best explains the relationship between X and Y
- Uses that line to predict Y for new X values

Example: S-H (X) vs Marks (Y) → 1→35, 2→45, 3→55, 4→65, 5→75, 6→85 → Linear pattern → LR draws a regression line → S-H=2.5 hours → O/P ≈ 50

LR Concept: Data points plotted on graph → Draw best fit straight line through them → Use line to predict new values

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2. Classification Algorithm

Used for predicting categories/labels.

- Output: Yes/No, Pass/Fail, Spam/Not Spam

Types of Classification Algorithms

- Logistic Regression ← very important
- K-Nearest Neighbors (KNN)
- Support Vector Machine (SVM)

- Decision Tree (DT)
- Random Forest (RF)

Logistic Regression

Despite its name, Logistic Regression is a Classification algorithm (not regression).

- Predicts categories (2 classes: Yes/No)
- Output is binary (0 or 1)

Example: Spam/Not Spam detection | Loan approval: Yes/No | Disease: Positive/Negative

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3. Algorithm Selection Summary

Problem Type	Algorithm Category	Examples
Predict a number	Regression	Linear Regression
Predict a category (2 classes)	Classification	Logistic Regression
Predict a category (multiple)	Classification	KNN, SVM, DT, RF

Day 4 20 April 2026 | Model Evaluation & Classification Metrics

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1. Choosing the Best Algorithm

Different algorithms give different results on the same data. We pick the best one using Model Evaluation.

> *Example: 5 Algorithms trained on same data → Alg 1→90%, Alg 2→94%, Alg 3→100%, Alg 4→98%, Alg 5→97% → Pick best*

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2. Model Evaluation

Process of measuring how well a machine learning model performs on data.

- To check how close predictions are to actual values
- Calculate error → Actual value − Predicted value

Two Types of Evaluation Metrics

Model Type	Metrics Used
Regression	MAE, MSE, RMSE, R^2 Score
Classification	Confusion Matrix, Accuracy, Precision, Recall

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3. Regression Metrics

Measure how close the predicted values are to the actual values.

i) MAE — Mean Absolute Error

Average of absolute errors (actual - predicted).

```
MAE = (1/n) × Σ |y - ŷ|
```

Actual (y)	Predicted (ŷ)	Error	Abs Error
10	8	2	2
20	18	2	2
30	33	-3	3

```
MAE = 7 / 3 = 2.33
```

- MAE = 0 → Perfect model
- Lower MAE = Better model

ii) MSE — Mean Squared Error

Average of squared errors. Penalizes large errors more.

```
MSE = (1/n) × Σ (y - ŷ)²
```

Actual	Predicted	Error	Squared Error
10	8	2	4
20	18	2	4
30	33	-3	9

```
MSE = 17 / 3 = 5.67
```

iii) RMSE — Root Mean Squared Error

Square root of MSE. Same unit as original values.

```
RMSE = √MSE = √5.67 ≈ 2.3
```

iv) R^2 Score — Coefficient of Determination

How well the model explains the data variance. Range: -∞ to 1.

```
R² = 1 - [ Σ(y - ŷ)² / Σ(y - ȳ)² ]
```

- y = Actual, ŷ = Predicted, ȳ = Mean of Actual
- 1 → Perfect prediction
- 0 → Model not useful
- < 0 → Worst model

Example: Calculated R^2 = 0.97 → Very Good (97% of variance explained)

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4. Classification Metrics

Measure how well a model predicts categories/classes.

Confusion Matrix

Evaluates correct and wrong predictions by comparing Actual vs Predicted labels.

	Predicted YES	Predicted NO
Actual YES	TP (True Positive)	FN (False Negative)
Actual NO	FP (False Positive)	TN (True Negative)

Definitions

- TP (True Positive) → Model predicted YES and it actually IS Yes
- TN (True Negative) → Model predicted NO and it actually IS No
- FP (False Positive) → Model predicted YES but actual is NO (Wrong +ve)
- FN (False Negative) → Model predicted NO but actual is YES (Wrong -ve)

Summary: Yes→Yes = TP | No→No = TN | Yes→No = FN | No→Yes = FP

Accuracy

How many predictions are correct out of total predictions.

```
Accuracy = (TP + TN) / (TP + TN + FP + FN)
```

Example: TP=2, TN=2, FP=1, FN=1 → Accuracy = 4/6 = 0.67 (67%)

Precision

Out of all predicted positive values, how many are actually correct.

```
Precision = TP / (TP + FP)
```

Example: TP=2, FP=1 → Precision = 2/3 = 0.67

Recall (Sensitivity)

Out of all actual positive values, how many did the model correctly identify.

```
Recall = TP / (TP + FN)
```

Example: TP=2, FN=1 → Recall = 2/3 = 0.67

Summary Table

Metric	Formula	Best Value
Accuracy	(TP+TN)/(TP+TN+FP+FN)	1.0 (100%)
Precision	TP/(TP+FP)	1.0
Recall	TP/(TP+FN)	1.0
MAE	$(1/n)\Sigma\lvert y-\hat{y}\rvert$	0 (no error)
R^2 Score	1 - SS_res/SS_tot	1.0

Day 5 21 April 2026 | Bias, Variance, Overfitting & Cross Validation

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1. Bias and Variance

Used to understand WHY a model performs well or poorly.

- Error in metrics → Find → Why it is performing that way → Bias and Variance

Bias

- Difference between actual values and predicted values
- Error caused when a model is too simple and cannot capture the true pattern
- High Bias → Underfitting
- Model ignores relationships in data → Wrong predictions

Example: Study hours 2,4,6,8,10 → Marks 20,40,60,80,100 → Model always predicts 60 regardless of hours → High Bias (MSE = 800)

Variance

- Error caused when a model is too complex and learns even the noise in the data
- Model learns too much, memorizes training data
- Very good on training data, poor on new data → High Variance
- High Variance → Overfitting

Underfitting (High Bias, Low Variance)

- Fails to learn patterns in the data
- Cannot capture the relationship in the data
- Training → Poor, Testing → Poor

Model too simple → Straight line when data is curved

Overfitting (Low Bias, High Variance)

- Too complex → learns even noise in the data
- When model learns training data too well but fails on new data
- Training → Good, Testing → Poor

Model memorizes training data → perfect on train but fails on unseen data

Ideal Model (Balanced)

Model State	Training Performance	Testing Performance
Underfitting (High Bias)	Poor	Poor
Overfitting (High Variance)	Very Good	Poor
Good/Balanced Model	Good	Good

High Bias → Underfitting | High Variance → Overfitting → Both cause errors → Need Balance

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2. Cross Validation

Evaluate a model more reliably by testing it on multiple different data splits.

- Train and test multiple times on different splits
- Reduces the impact of underfitting and overfitting

K-Fold Cross Validation

- Dataset divided into K equal parts (folds)
- Model is trained and tested K times
- Each fold gets a turn as the test set

K=5 → 5-fold cross validation → Model trains 5 times, each time different fold is test set

Example: K=5 with dataset [(1,10),(2,20),(3,30),(4,40),(5,50)]

Iteration	Train Data	Test Data
1	(2,20),(3,30),(4,40),(5,50)	(1,10)
2	(1,10),(3,30),(4,40),(5,50)	(2,20)
3	(1,10),(2,20),(4,40),(5,50)	(3,30)
4	(1,10),(2,20),(3,30),(5,50)	(4,40)
5	(1,10),(2,20),(3,30),(4,40)	(5,50)

- Each and every record trained 4 times → Avoids Underfitting and Overfitting
- K=100 → randomly selected (LOO - Leave One Out)

Effect of K Value

K Value	Effect	Train	Test
K=1	High Variance (Overfitting)	Very Good	Bad
K=5	High Bias (Underfitting)	Bad	Bad
K=3	Good Model [Balanced]	Good	Good

Hyperparameter Tuning: Try different K values on dataset and select the one that gives best results (using Grid Search or Randomized Search)

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3. ML Pipeline with Cross Validation

Data → Algorithm → Bias/Variance check → Metrics → Predict/CV → Output

- VS Code → Deploy → 1 (production)

Day 6 22 April 2026 | Hyperparameter Tuning & NLP Introduction

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1. Hyperparameter Tuning

Process of finding the best parameters for an ML model before training it. Model gives best result.

- Hyperparameter = setting we choose before training (like K in K-Fold)
- Regular parameter = learned during training

Cross Validation → K=5 → Try K=1,2,3,4,5 → Find best K → Hyperparameter Tuning

Hyperparameter Tuning Methods

- Grid Search → Try all combinations: K=1, K=2, K=3, K=4, K=5 systematically
- Randomized Search → Try random combinations: K=3, K=6, K=2

Effect of K on Model

K Value	Effect	Train Result	Test Result
K=1	High Variance (Overfitting)	Good	Bad
K=5	High Bias (Underfitting)	Bad	Bad
K=3	Balanced (Best Model)	Good	Good

Best Practice: Try different K values → Split → Regression/Classification Metrics → Accurate Results

Full Pipeline with Hyperparameter Tuning

Data → DP → Algorithm → (HPT) → Model Evaluation → New Data → Output

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2. NLP — Natural Language Processing

Field of AI that helps computers understand, read, and process human language.

- Text → Numbers (ML model understands only numbers)

Examples: Chatbot, Google Translate | Hello → Machine → Hi, How are you?

Text Preprocessing

Process of cleaning and preparing text data before giving it to an ML model.

Raw text example: 'Hello!!! I am Learning Data Science!!!'

Steps of Text Preprocessing

- 1. Lowercase conversion → Convert all text to small letters (lowercasing)
- 2. Remove Punctuation → Hello!!! → Hello
- 3. Remove Stop Words → I, am, the, is, are etc → filtered out
- 'hello i am learning data science' → 'hello learning datascience'
- 4. Convert words to base form → Stemming / Lemmatization
- learning → learn | running → run

Cleaned text: 'hello learn data science' → tokens passed to machine as numbers

Goal of Text Preprocessing

Action	Purpose
Remove noise	Cleaner input for model
Reduce data size	Faster processing
Make data consistent	Uniform format

Day 7 23 April 2026 | NLP: Vectorization — BoW & TF-IDF

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1. Vectorization in NLP

Process of converting text data into numerical format so ML algorithms can process it.

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2. Bag of Words (BoW)

- Count how many times each word appears in the document
- Ignore grammar and word order

Sentences: S1 = 'I love NLP' | S2 = 'I love AI' → Vocabulary = ['I','Love','NLP','AI']

Sentence	I	love	NLP	AI
S1 (I love NLP)	1	1	1	0
S2 (I love AI)	1	1	0	1

- Importance → represented as 0 or 1 (presence/absence)

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3. TF-IDF — Term Frequency-Inverse Document Frequency

Converts text into numerical values by giving importance to words in a document.

TF — Term Frequency

Measures how frequently a word appears in one document.

```
TF(t,d) = (No. of times term t appears in document) / (Total no. of terms in document)
```

- If word appears more → Higher TF

IDF — Inverse Document Frequency

Measures how important a word is across all documents.

```
IDF(t) = log( N / (1 + DF(t)) )
```

- N → total number of documents
- DF(t) → number of documents containing the word
- Common words (like 'the', 'a') → low IDF
- Rare important words → high IDF

TF-IDF Score

```
TF-IDF(t,d) = TF(t,d) × IDF(t)
```

Higher TF-IDF = more important word in that document across the corpus

Application: Spam Email Detection

- 1. Collect Name & email dataset
- 2. Upload dataset → Label: Spam(1) / Not Spam(0)

- 3. Filter → Vectorize → Machine learns patterns
- Data → LR → NLP → New data → Output (Spam/Not Spam)

Day 8 24 April 2026 | ML Deployment Pipeline

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1. ML Deployment Pipeline

Converting raw text data into meaningful predictions — end-to-end system.

Complete ML Pipeline Steps

- 1. Problem Understanding — Define what to solve
- 2. Data Collection — Gather relevant data
- 3. Data Preprocessing — EDA (Exploratory Data Analysis)
- 4. Label Encoding — Convert categories to numbers
- 5. NLP Preprocessing — Clean and prepare text
- 6. Vectorization — Convert text to numbers (BoW / TF-IDF)
- 7. Train/Test Split — 80% Train, 20% Test
- Train → Learn | Test → Evaluate
- 8. Model Building — Choose Algorithm (e.g., Logistic Regression)
- 9. Model Evaluation
- Accuracy → 84% to 95%
- Confusion Matrix
- 10. Cross Validation — Train model multiple times → Reduce bias
- 11. Deployment (Streamlit) — Build web app, enter review, get output

Machine Learning = Data + Processing + Model + Evaluation + Deployment

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2. Capstone Formula

The complete definition of Machine Learning as a system:

Data + Processing + Model + Evaluation + Deployment

= Machine Learning

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3. Projects (NLP — Suggested Assignments)

Build and post on LinkedIn (tag instructors):

- 1. Sentiment Analysis App (Movie/Product Reviews) → Classify as Positive or Negative
- 2. Fake News Detection System → Detect Fake / Real news
- 3. Resume Classification System → Classify into Data Science, Web Development, Marketing
- 4. Customer Support Ticket Classifier → Billing, Technical Issues, General Query
- 5. Toxic Comment Detection → Social Media Comments → Toxic / Safe

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4. Quick Reference — All Concepts

Day	Topic	Key Concepts
1	Data Analysis & ML Intro	DA vs ML, AI, Supervised/Unsupervised, Regression, Classification
2	Data Preprocessing	Cleaning, Encoding, Scaling (Normalization & Standardization), Splitting
3	Regression Algorithms	Linear Regression, Logistic Regression, Classification Algorithms
4	Model Evaluation	MAE, MSE, RMSE, R^2, Confusion Matrix, Accuracy, Precision, Recall
5	Bias & Variance	Underfitting, Overfitting, Cross Validation, K-Fold
6	Hyperparameter Tuning + NLP	Grid Search, Randomized Search, Text Preprocessing
7	NLP Vectorization	Bag of Words (BoW), TF-IDF, TF formula, IDF formula
8	ML Deployment Pipeline	Full pipeline: Collect → Preprocess → Model → Evaluate → Deploy